

Experiment No. 9

Title

Understanding INodes, I/O Redirection, Piping and Process Control Commands in Linux

Aim

The problem is to understand the concept of inodes, practice I/O redirection and piping, and use process control commands to manage running processes in a Linux environment.

Theory

Inodes

An inode is a data structure that stores metadata of a file such as:

- File size
- Ownership
- Permissions
- Timestamps
- Disk location

I/O Redirection

Redirection allows command outputs to be redirected to files instead of the terminal:

- > – Redirect output (overwrite)
- >> – Redirect output (append)
- < – Redirect input
- 2> – Redirect error output Pipes (|) allow the output of one command to be used as input to another command. Example: `ls | grep txt`

Pipes

Pipes (|) allow the output of one command to be used as input to another command.
Example: `ls | grep txt`

Process Control

Linux supports multitasking. Processes can be monitored and controlled using commands such as:

- ps
- top
- kill
- jobs
- bg, fg

Commands Used

Commands	Description
ls -i	Display inode number of files
stat	Display file inode details
>, >>	Output redirection
<	Input redirection
ps	Display running processes
top	Real-time process monitoring
kill	Terminate a process
jobs	List background jobs
bg	Resume job in background

Procedure / Steps

Step 1: Understanding Inodes

1. Create a test file: `touch inode_test.txt`
2. Display inode number: `ls -li inode_test.txt`
3. View inode details: `stat inode_test.txt`

```
(amanyadav@kali-vm)-[~]  
$ touch inode_test.txt  
  
(amanyadav@kali-vm)-[~]  
$ ls -li inode_test.txt  
2217 inode_test.txt  
  
(amanyadav@kali-vm)-[~]  
$ stat inode_test.txt  
File: inode_test.txt  
Size: 0          Blocks: 0          IO Block: 4096   regular empty file  
Device: 8,1      Inode: 2217       Links: 1  
Access: (0664/-rw-rw-r--)  Uid: ( 1000/amanyadav)   Gid: ( 1000/amanyadav)  
Access: 2026-04-27 18:12:40.250669561 +0530  
Modify: 2026-04-27 18:12:40.250669561 +0530  
Change: 2026-04-27 18:12:40.250669561 +0530  
Birth: 2026-04-27 18:12:40.238089106 +0530
```

Step 2: Input / Output Redirection

4. Redirect output to a file: `ls > file_list.txt`
5. Append output: `date >> file_list.txt`
6. Redirect input: `wc -l < file_list.txt`

```
(amanyadav@kali-vm)-[~]  
$ ls > file_list.txt  
  
(amanyadav@kali-vm)-[~]  
$ date >> file_list.txt  
  
(amanyadav@kali-vm)-[~]  
$ wc -l < file_list.txt  
16
```

Step 3: Using Pipes

7. Combine commands using pipe: `ls -l | grep ".txt"`
8. Count number of text files: `ls | grep ".txt" | wc -l`

```
(amanyadav@kali-vm)-[~]
$ ls -l | grep ".txt"
-rw-rw-r-- 1 amanyadav amanyadav 179 Apr 27 18:15 file_list.txt
-rw-rw-r-- 1 amanyadav amanyadav  0 Apr 27 18:12 inode_test.txt

(amanyadav@kali-vm)-[~]
$ ls | grep ".txt" | wc -l
2
```

Step 4: Process Control Commands

9. Display running processes: ps
10. Monitor processes in real-time: top
11. Run a command in background: sleep 60 &
12. List background jobs: jobs
13. Bring job to foreground: fg
14. Terminate a process: kill<PID>

```
(amanyadav@kali-vm)-[~]
$ ps
  PID TTY          TIME CMD
 1681 pts/0        00:00:00 bash
 5248 pts/0        00:00:00 ps
```

```
(amanyadav@kali-vm)-[~]
$ top
top - 18:19:11 up 8 min,  1 user,  load average: 0.53, 0.55, 0.37
Tasks: 178 total,  1 running, 176 sleeping,  0 stopped,  1 zombie
%Cpu(s):  2.3 us,  7.8 sy,  0.0 ni, 88.6 id,  0.0 wa,  0.0 hi,  1.4 si,  0.0 st
MiB Mem : 4632.9 total, 3111.8 free,  903.6 used,  842.3 buff/cache
MiB Swap: 480.0 total,  480.0 free,  0.0 used. 3729.3 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
    596 root        20   0 418904 138868 82856 S   6.6   2.9   0:25.86 Xorg
    897 amanyad+   9  -11 176480 16504  8920 S   5.0   0.3   0:08.78 pipewire-pulse
    892 amanyad+   9  -11 105356 15496  9452 S   3.6   0.3   0:09.57 pipewire
   1077 amanyad+  20   0 624888 142164 99760 S   3.6   3.0   0:06.44 xfwm4
   1000 amanyad+  20   0 281152  3372  2960 S   1.0   0.1   0:05.09 VBoxClient
   1137 amanyad+  20   0 296040 48396 21460 S   1.0   1.0   0:05.54 wrapper-2.0
   1139 amanyad+  20   0 272252 29056 21968 S   1.0   0.6   0:04.73 wrapper-2.0
    486 message+ 20   0  10228  7136  4300 S   0.3   0.2   0:01.43 dbus-daemon
    493 root        20   0  18260  9016  7680 S   0.3   0.2   0:00.53 systemd-logind
    554 root        20   0 353140  2996  2728 S   0.3   0.1   0:02.07 VBoxDRMClient
   1067 amanyad+  20   0  90160  3992  3684 S   0.3   0.1   0:00.04 gpg-agent
   1142 amanyad+  20   0 360260 40368 32236 S   0.3   0.9   0:00.44 wrapper-2.0
   1184 root        20   0 318828 10736  8908 S   0.3   0.2   0:01.03 upowerd
   1652 amanyad+  20   0 654748 66288 50780 S   0.3   1.4   0:08.07 qterminal
      1 root        20   0  24608 15088 11364 S   0.0   0.3   0:01.94 systemd
      2 root        20   0      0      0      0 S   0.0   0.0   0:00.02 kthreadd
      3 root        20   0      0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root        0 -20      0      0      0 I   0.0   0.0   0:00.13 kworker/0:0H-kblockd
```

```
(amanyadav@kali-vm)-[~]  
$ sleep 60 &  
[1] 6407  
  
(amanyadav@kali-vm)-[~]  
$ jobs  
[1]+  Running                  sleep 60 &  
  
(amanyadav@kali-vm)-[~]  
$ fg  
sleep 60
```

Outcomes

- ✓ Inode numbers and file metadata displayed correctly.
- ✓ Command output redirected and appended successfully.
- ✓ Pipes used to filter and process output.
- ✓ Processes monitored and controlled successfully.

Result

Inode information, input/output redirection, piping, and process control commands were successfully executed. The student gained practical understanding of Linux file system internals and process management.